

Chapter 5

ALKENES

**STRUCTURES, NOMENCLATURE, AND AN INTRODUCTION TO
REACTIVITY
THERMODYNAMICS AND KINETICS**

Alkenes

- Saturated compounds (alkanes) have the maximum number of hydrogen atoms attached to each carbon atom
- Unsaturated compounds have fewer hydrogen atoms attached to the carbon chain than alkanes
- Unsaturated compounds contain double or triple bonds

Properties of Alkenes

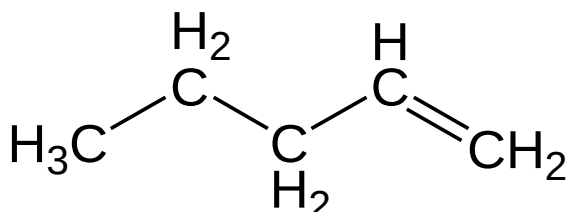
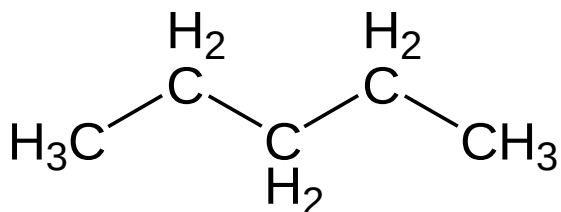
- Alkenes are hydrocarbons containing at least one double bond



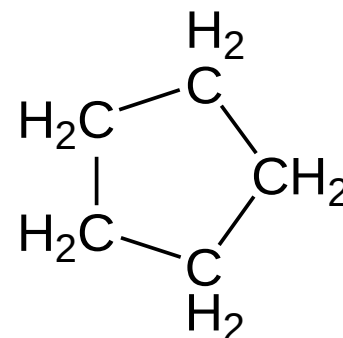
- Alkenes are compounds with general formula (C_nH_{2n})
- Degree of unsaturation = no. of π -bonds and rings

1 double bond is equal to **1** degree of saturation
1 ring is equal to **1** degree of saturation
1 triple bond is equal to **2** degree of saturation

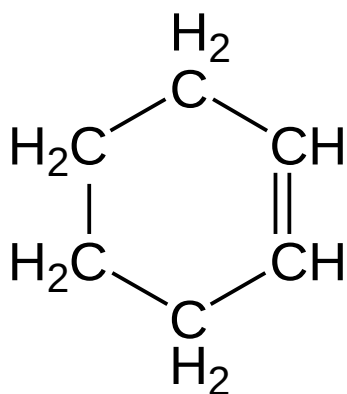
Molecular Formula



1 degree of
unsaturation



1 degree of
unsaturation



$$2n+2 = 2 \times 6 + 2 = 14 \text{ hydrogens}$$

$$14 - 10 = 4$$

$$4/2 = 2 \text{ degree of unsaturation}$$

**Q: An unknown hydrocarbon have 9 carbons, 2 rings, 1 double bond and 1 triple bond.
What is the molecular formula?**



is the molecular formula for alkane based on no. of carbon.

Degree of saturation = 5

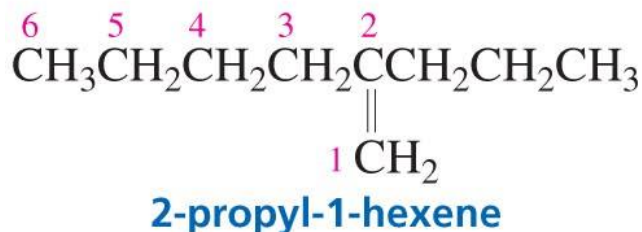
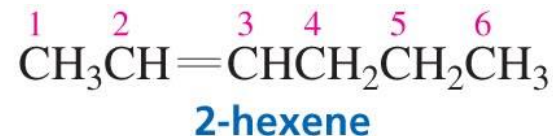
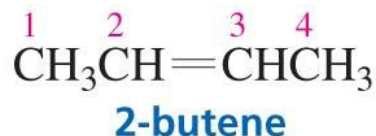
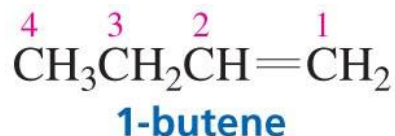
(2 ring + 1 double bond+ 2 triple)

$5 \times 2 = 10$ (no. of missing H)

So, molecular formula = C_9H_{10}

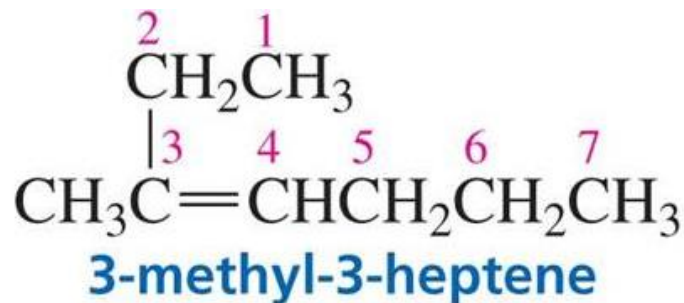
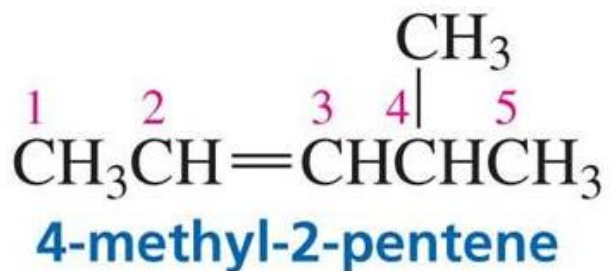
Nomenclature of Alkenes

1. Longest continuous chain containing the **functional group**:
2. Number the chain so that the **functional group** gets the lowest number

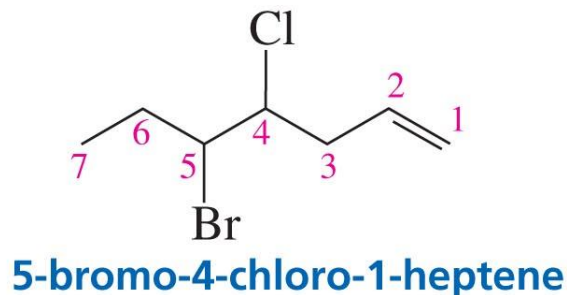
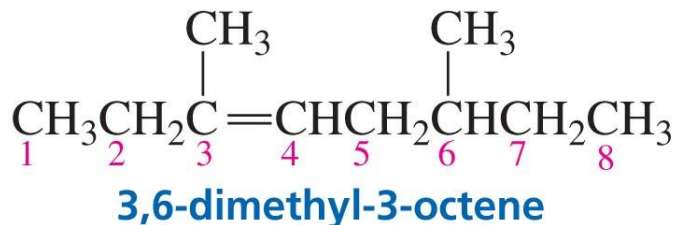


the longest continuous chain has eight carbons but the longest continuous chain containing the functional group has six carbons, so the parent name of the compound is hexene

3. When there are both a functional group and a substituent, the functional group gets the lowest number:

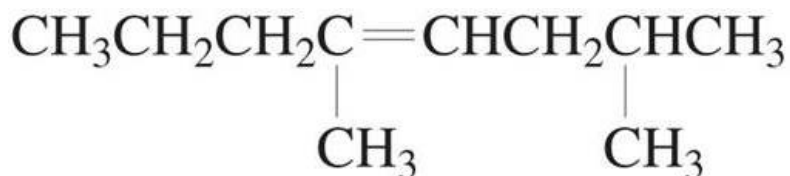


4. Cite the substituents in alphabetical order:



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5. Name with the lowest functional group number and then the lowest substituent numbers:

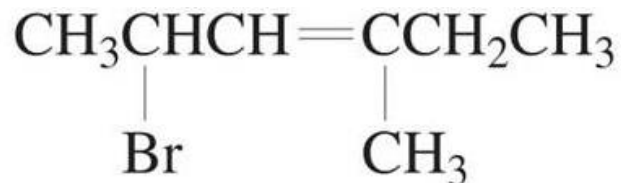


2,5-dimethyl-4-octene

not

4,7-dimethyl-4-octene

because $2 < 4$



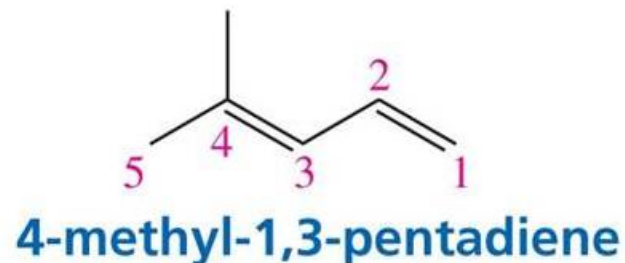
2-bromo-4-methyl-3-hexene

not

5-bromo-3-methyl-3-hexene

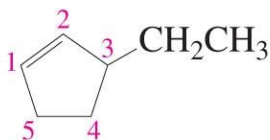
because $2 < 3$

6. For a compound with two double bonds, the suffix is “diene”:



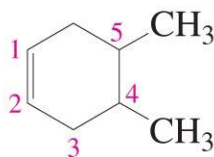
Nomenclature of Cycloalkenes (C_nH_{2n-2})

1. In a cyclic alkene, the alkene functional group is always given the numbers 1 and 2.

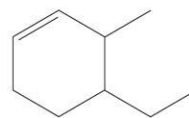


3-ethylcyclopentene

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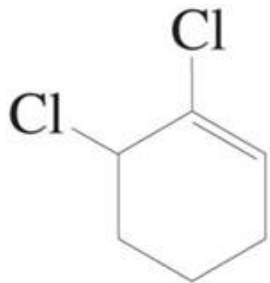


4,5-dimethylcyclohexene



4-ethyl-3-methylcyclohexene

2. The alkene substituent is always given the number 1

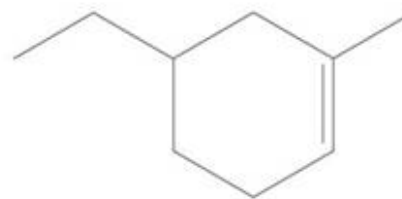


1,6-dichlorocyclohexene

not

2,3-dichlorocyclohexene

because $1 < 2$



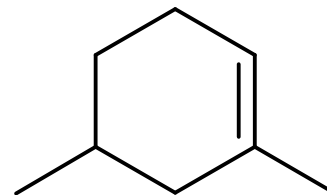
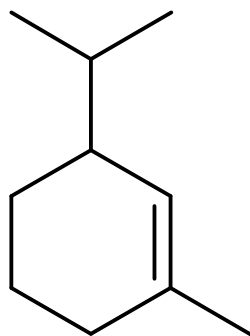
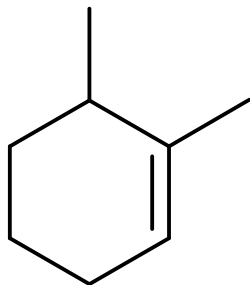
5-ethyl-1-methylcyclohexene

not

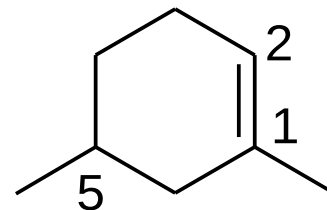
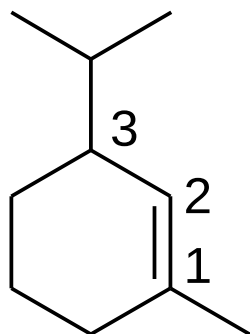
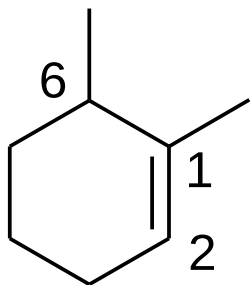
4-ethyl-2-methylcyclohexene

because $1 < 2$

Q: What is the IUPAC name of these cycloalkenes?



Answer:



3-isopropyl-1-methylcyclohexene

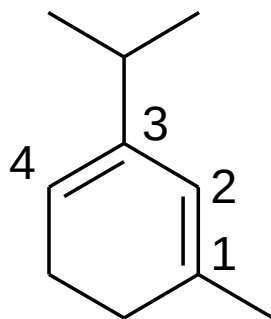
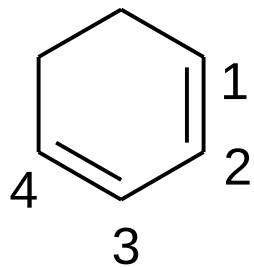
1,6-dimethylcyclohexene

1,5-dimethylcyclohexene

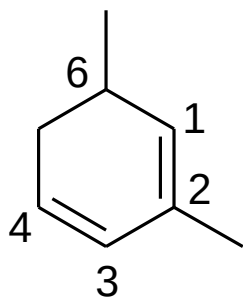
Nomenclature of Cycloalkenes with two or more double bonds

Cycloalkadiene (C_nH_{2n-4})

1. In a cyclic diene, the diene functional groups are always given the least numbers and with numbers 1 and 2 on the first double bond.

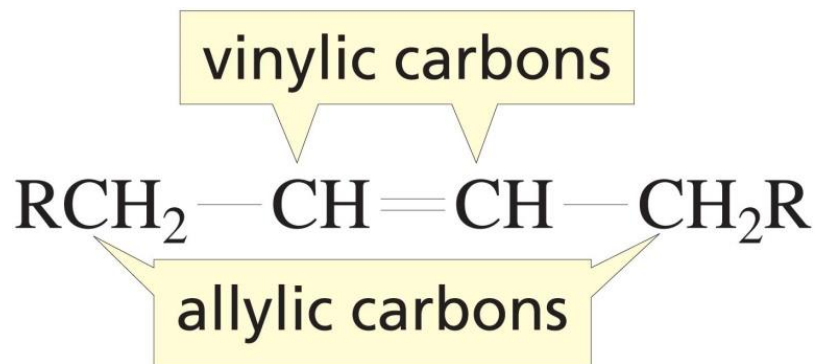


1,3-cyclohexadiene

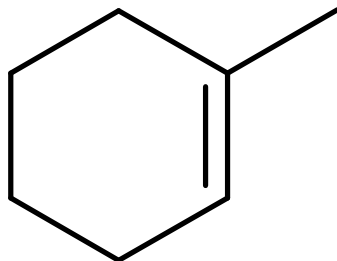


2,6-dimethylcyclo-1,3-hexadiene

Special Nomenclatures



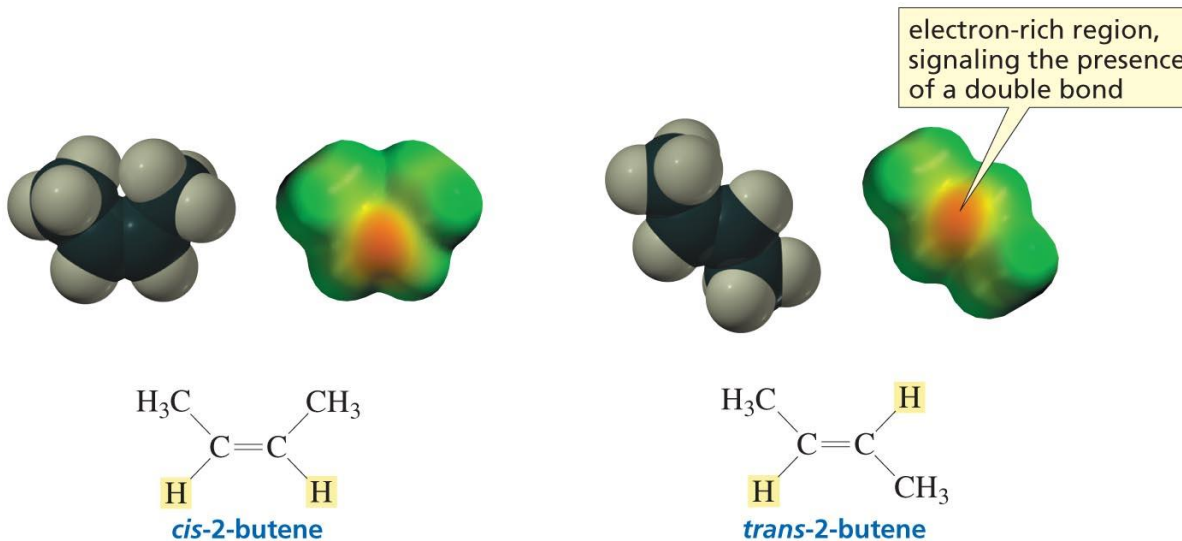
Q: How many 2° allylic carbons are in methylcyclohexene?



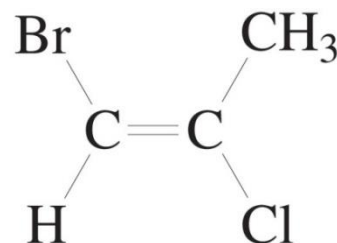
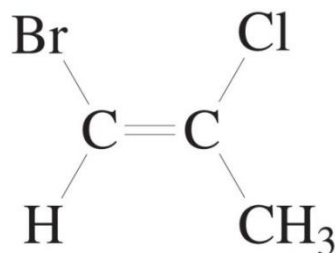
Two secondary allylic carbons

Alkene Isomers

(Cis-Trans VS E-Z relationships)



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Which isomer is cis and which is trans?

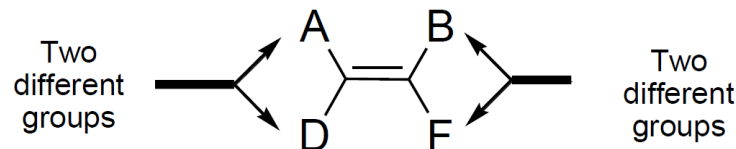
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Alkene Isomers

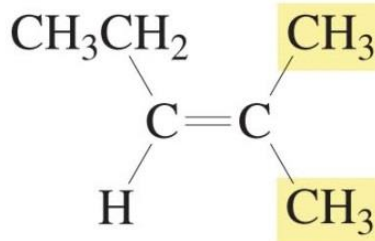
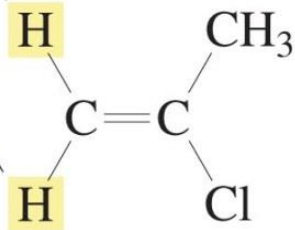
(Cis-Trans VS E-Z relationships)

One condition:

Each sp^2 carbon should have at least two different groups.

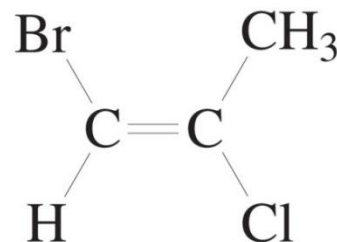
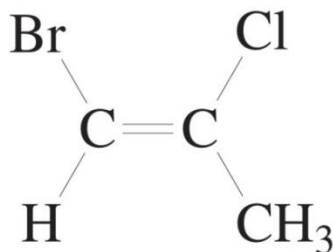


cis and trans isomers are not possible for these compounds because two substituents on an sp^2 carbon are the same



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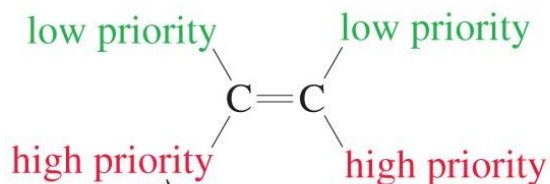
Is this Cis or Trans??



Which isomer is cis and which is trans?

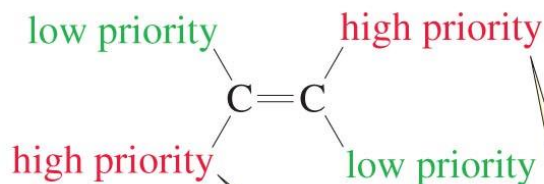
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E and *Z* Nomenclature:



the *Z* isomer has the high priority groups on the same side of the double bond

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the *E* isomer has the high priority groups on opposite sides of the double bond

E, Z is from German:

***Z*, Zusammen**
(Means together)

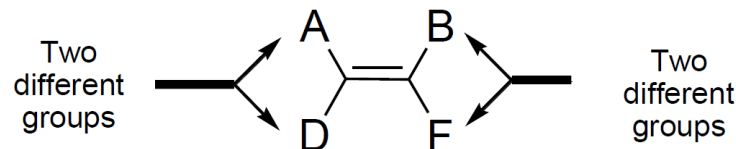
***E*, Entgegen**
(means opposite)

Alkene Isomers

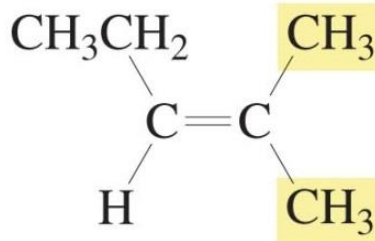
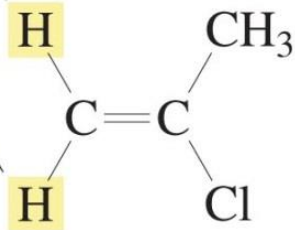
(Cis-Trans VS E-Z relationships)

One condition:

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cis and trans isomers are not possible for these compounds because two substituents on an sp^2 carbon are the same

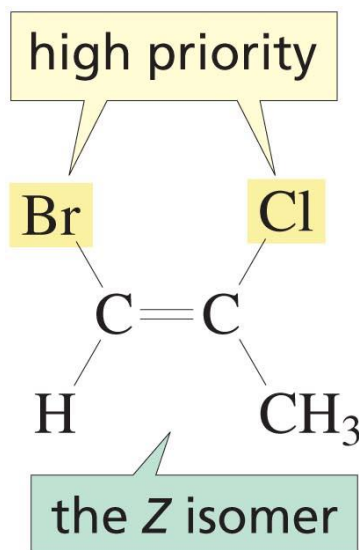


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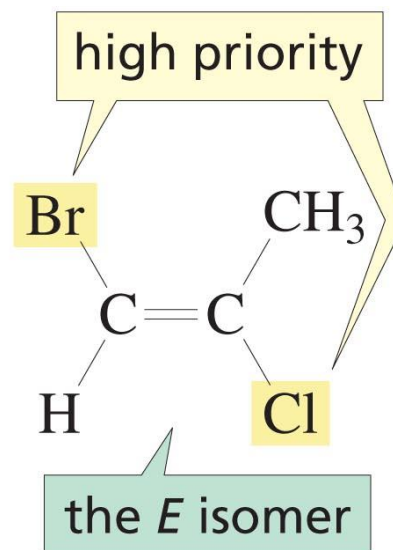
Naming by the *E,Z* System

Rule 1: Consider the atomic number of the atoms bonded directly to a specific sp^2 carbon:

53 35 17 16 8 7 6 1
I > Br > Cl > S > O > N > C > H

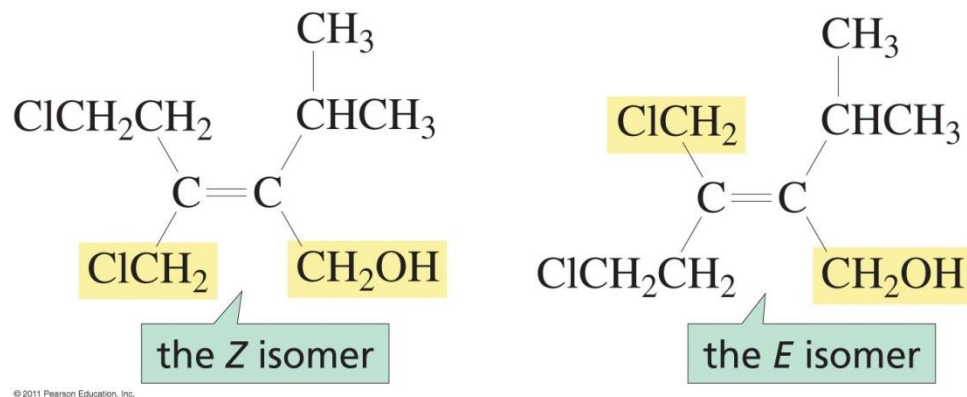


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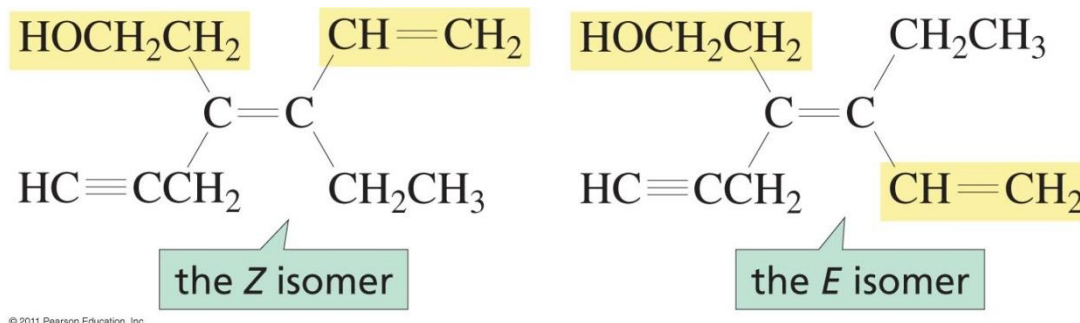


Higher atomic number has higher priority

Rule 2: If there is a tie, consider the atoms attached to the tied atoms:

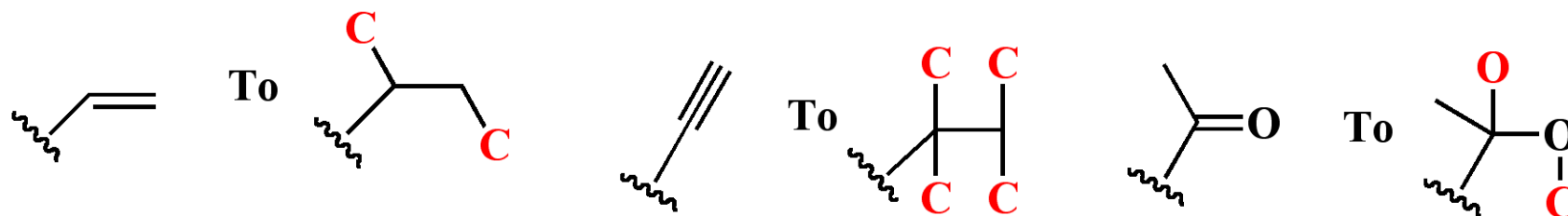


Rule 3: Multiple bonds are treated as attachment of multiple single bonds using “divide-duplicate.”

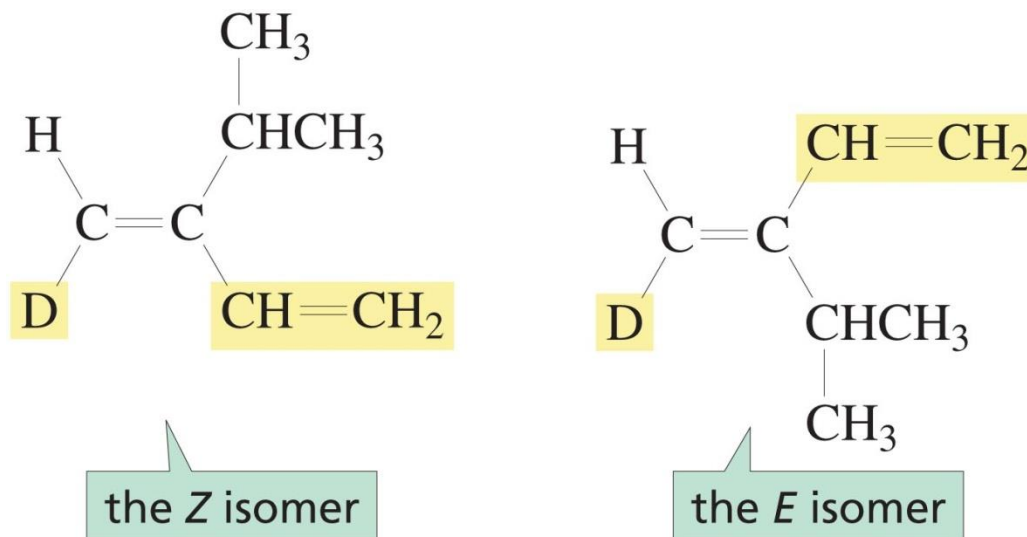


Using divide-duplicate:

- Divide π -bonds.
- Duplicate atoms attached to each π -bond.

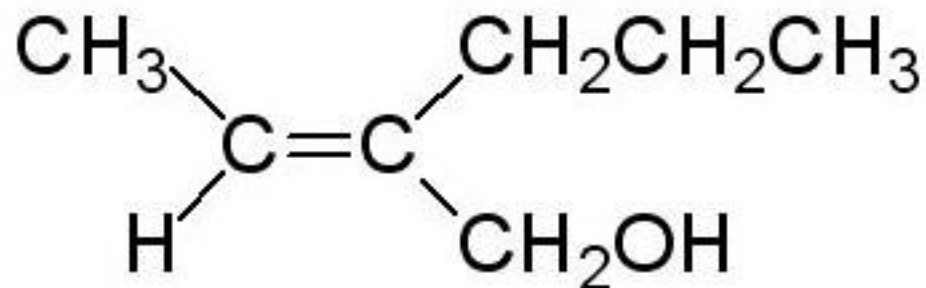
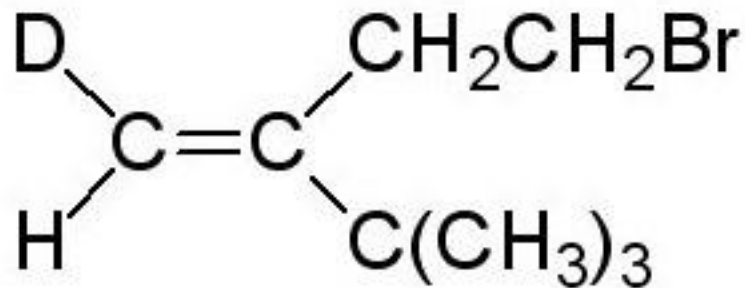


Rule 4: Rank the priorities by mass number in isotopes:



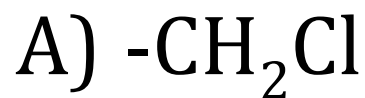
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Q: Assign the configuration of the alkenes below as Z or E, respectively.



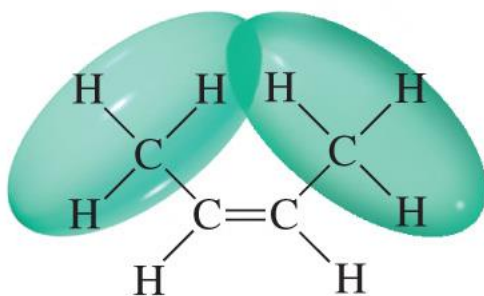
- A) E,E
- B) E,Z
- C) Z,Z
- D) Z,E

Q: Which one of the following groups has the highest rank ?



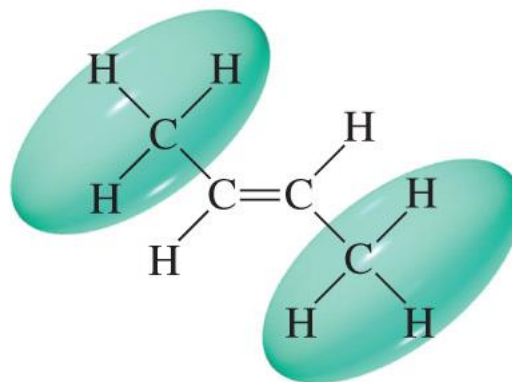
Alkene Stability

the cis isomer has steric strain



cis-2-butene

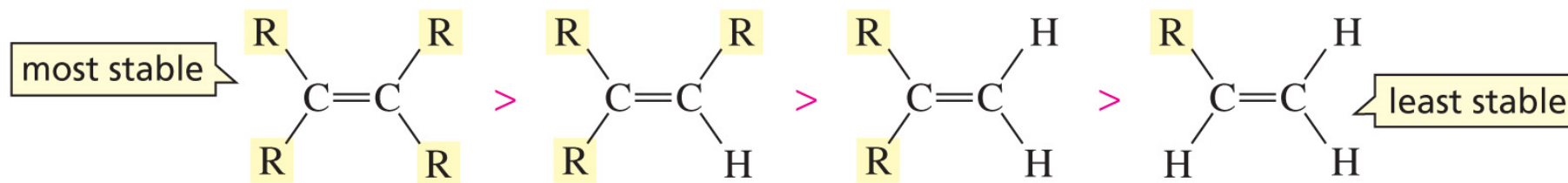
the trans isomer does not have steric strain



trans-2-butene

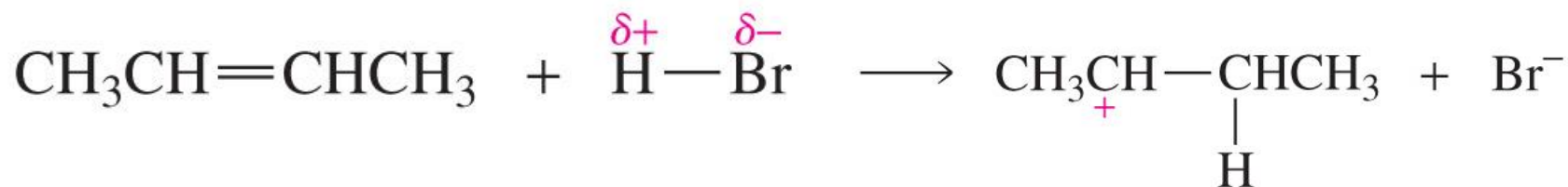
Alkene Stability

relative stabilities of alkyl-substituted alkenes

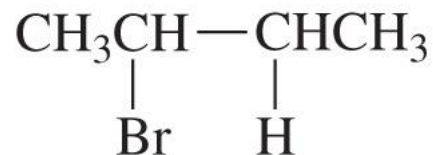


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Reactions

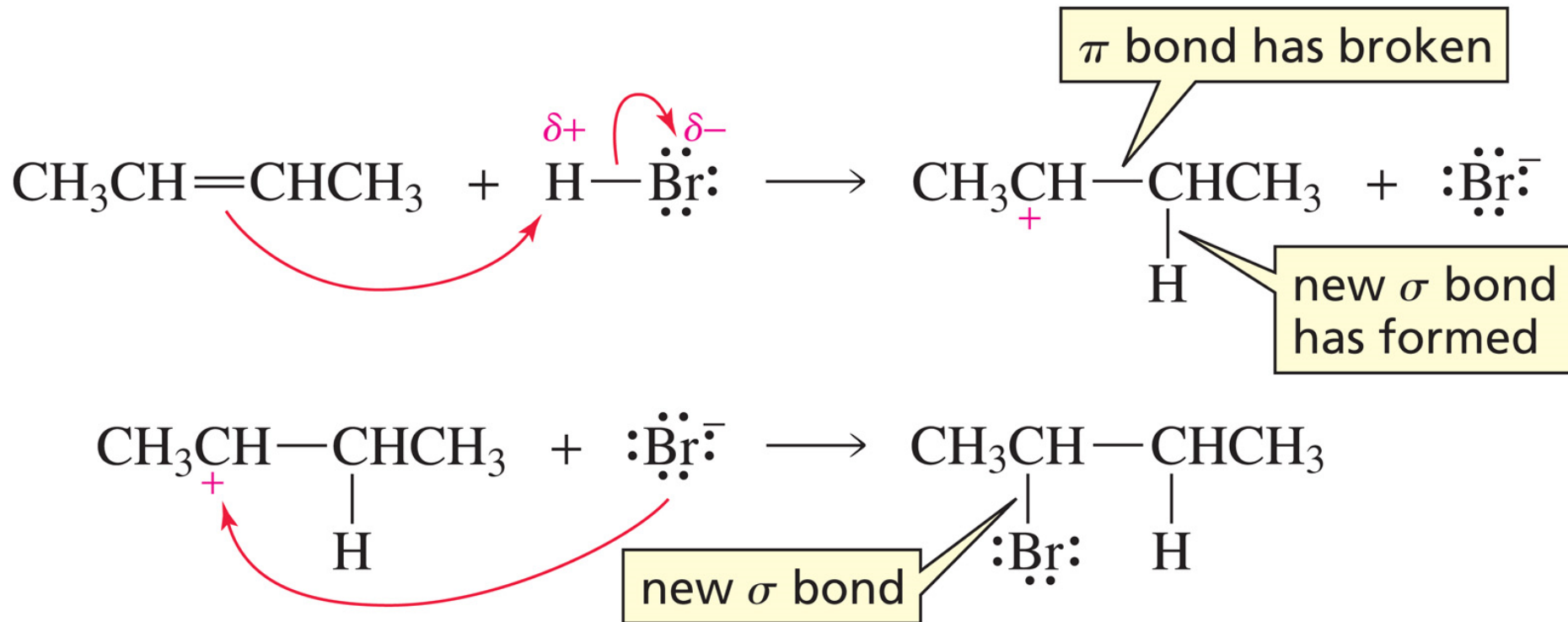


a carbocation



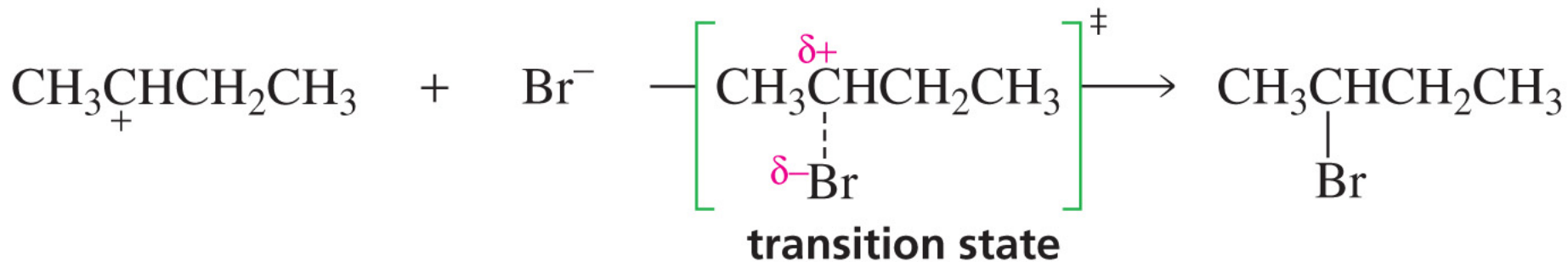
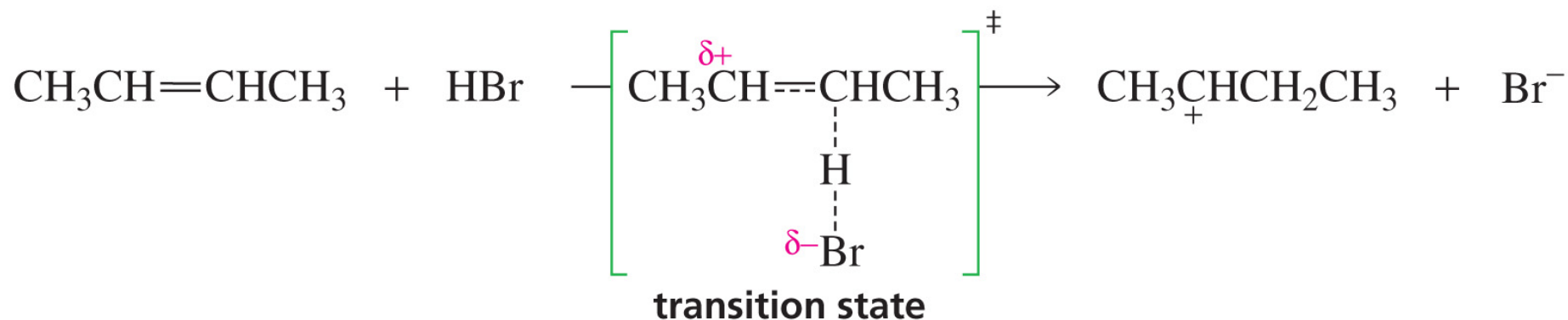
2-bromobutane
an alkyl halide

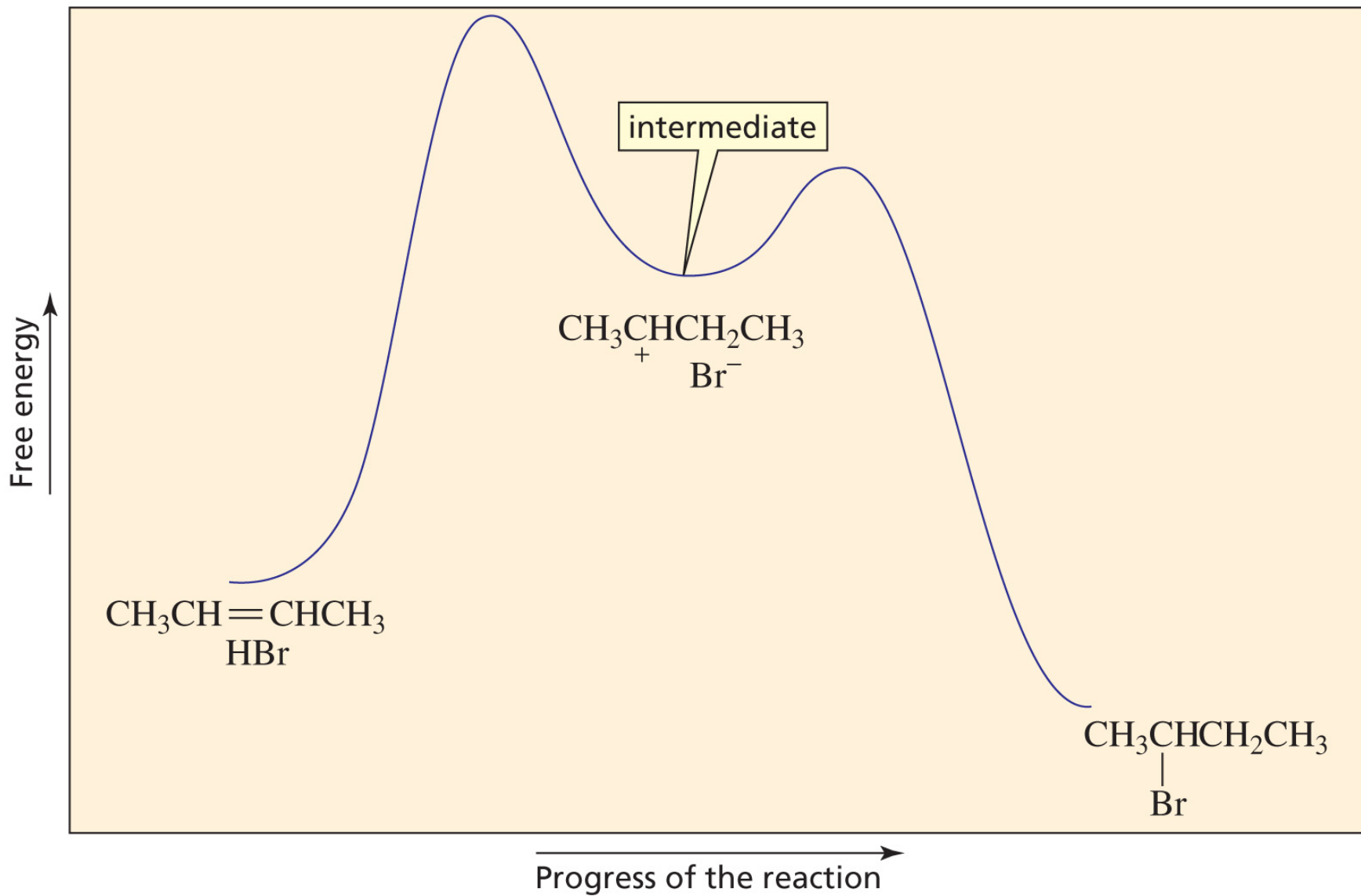
Mechanism



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Mechanism





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Energy Changes in a Reaction

- Enthalpy changes, ΔH^0 , for a reaction arises from changes in bonding in the molecule.
 - If weaker bonds are broken and stronger ones formed then ΔH^0 is negative and *exothermic*.
 - If stronger bonds are broken and weaker ones formed then ΔH^0 is positive and *endothermic*.

Gibbs Free Energy

$$\Delta G^0: \quad \Delta G^0 = \Delta H^0 - T\Delta S^0$$

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger \quad (\text{Free energy of activation})$$

Reaction Rate

Reaction Rate depends on

- 1- number of collisions
- 2- fraction with sufficient energy
- 3- fraction with proper orientation

Catalyst: increase the rate of reaction by decreasing the energy barrier

